

YOGENDRA GUPTA

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PROFESSIONAL SUMMARY

Computer Science undergraduate at VIT Vellore (Class of 2027) with hands-on experience building and deploying backend and full-stack applications using Python, Java, and JavaScript/TypeScript. Experienced in designing FASTAPIs, implementing authentication and database-driven systems, and deploying applications on Linux-based environments. Strong interest in backend engineering and AI/ML, with a practical approach to building, debugging, securing, and deploying end-to-end systems.

EDUCATION

VIT Vellore — B.Tech in Computer Science Engineering

2023 – 2027 | CGPA: 8.14

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Cloud Architecture, Cloud Computing, Data Privacy, Web Application Security

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript

Backend: FastAPI, REST APIs, JWT Authentication & Authorization, SQLAlchemy ORM

Frontend: React, TypeScript, Tkinter

Databases: SQLite, Database Design

Cloud & DevOps: AWS EC2, Nginx, Linux (Ubuntu), Vercel, Render

AI / ML: OpenCV, Object Detection, NumPy, Pandas

Tools: Git, GitHub

Core CS: Data Structures & Algorithms, OOP, System Design Fundamentals

PROJECTS

FastAPI Backend Service | [GitHub](#) | [API Docs](#)

Production-grade backend that authenticates users and serves modular business logic on a live cloud deployment.

- Architected a layered FastAPI service (routers → services → data-access) to isolate auth, business logic, and persistence.
- Replaced raw SQL with SQLAlchemy ORM, closing off SQL-injections, then deployed and hardened it on AWS EC2 (Ubuntu) behind an Nginx reverse proxy running Uvicorn.

Manual JWT Notes App | [GitHub](#) | [Live Demo](#)

Secured notes API implementing JWT authentication from first principles instead of a higher-level auth library.

- Hand-built the HMAC-SHA256 token pipeline — signing, encoding, expiry validation, signature verification — wired into FastAPI's dependency-injection system to gate every protected route.
- Hashed credentials with bcrypt via Passlib, modeled users/notes with SQLAlchemy ORM against SQLite, and delivered full note CRUD plus account deletion behind Bearer-token verification.

Jumbled Frames Reconstruction Challenge | [GitHub](#) | [Demo](#)

Video-reordering pipeline that reconstructs a shuffled 300-frame clip back into its original temporal sequence.

- Combined SSIM and HSV histogram similarity with a windowed comparison strategy to reorder frames via greedy nearest-neighbor search, parallelizing computation with multiprocessing.
- Exposed Fast/Balanced/Accurate modes to trade off runtime against reconstruction accuracy.

YOLO Playground — Object Detection App | [GitHub](#)

End-to-end computer vision pipeline that turns raw images into real-time, confidence-scored detections.

- Designed a YOLO + OpenCV inference pipeline with tuned confidence thresholds, plus a Streamlit interface for image upload and result visualization.

Chrome Highlight Extension | [GitHub](#)

Lightweight browser extension that captures and syncs highlighted web content without a dedicated backend.

- Built a Chrome extension with JavaScript and the Extension APIs to capture highlighted text with source URL/timestamp metadata, syncing it to Google Sheets via Apps Script as a serverless backend.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — Oracle — [View](#)
- Generative AI — IBM — [View](#)
- Python Object-Oriented Programming — LinkedIn Learning — [View](#)
- System Design in Cloud — LinkedIn Learning — [View](#)
- Introduction to Data Science — Infosys Springboard — [View](#)

HACKATHONS

- Yantra'25 — Code4Change 2.0 Hackathon